Where to look first (key classes and functions)

What	Where	Why it matters
Trained agent (the ML model)	`src/hex_ml/agents/trained_agent.py` -> `TrainedHexAgent`, `mask_illegal_moves`	Loads the trained weights, encodes the board, predicts a move, masks illegal cells. This is the delivered agent.
Policy network architecture	`src/hex_ml/training/model.py` -> `TinyHexPolicyNet`	The actual neural network: board in, per-cell move scores out.
State encoding	`src/hex_ml/encoding/board_encoder.py` -> `encode_board_channels`	Defines what the network sees (4-channel board state).
Training loop	`scripts/train_11x11.py` -> `main`	Where imitation learning happens: loads records, trains `TinyHexPolicyNet` with cross-entropy against expert moves.
Dataset construction	`src/hex_ml/data/dataset_builder.py` -> `imitation_records_from_game_log`	Turns a raw game log into (state, expert action) training pairs.
Leakage-free splitting	`src/hex_ml/data/splitting.py` -> `split_game_ids`, `assert_no_game_id_leakage`	Ensures train/validation/test never share a game.
External/expert agent adapter	`src/hex_ml/agents/gamelib_wrapper.py` -> `GamelibAgentWrapper`	Generic loader so any hackathon-style agent (expert or opponent) can be called the same way.
Expert agent specifically	`src/hex_ml/agents/expert_wrapper.py` -> `ExpertAgentWrapper`	The winner agent (Adyansh Gupta) used only to generate/label training data.
Opponent auto-discovery	`src/hex_ml/agents/discovery.py` -> `discover_gamelib_agents`	Finds every agent under `external_agents/` automatically.
Data generation with structured opponents	`scripts/generate_structured_opponent_dataset.py`	Explains and implements why algorithm-based opponents were used instead of random play.
Combining datasets	`scripts/build_combined_dataset.py`	Merges the structured and DAgger-round datasets with traceable provenance.
Phase-based evaluation	`scripts/analyze_phase_performance.py` -> `phase_from_record`	Breaks down accuracy by opening/midgame/endgame.
Tests	`tests/`	37 tests covering board rules, encoding, splitting/leakage, and the gamelib orientation fix.